



CS5071NI - Professional and Ethical Issues

2025-26 Autumn

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Figure 1 : Dr. Charlie Miller and Chris Valasek

1. Introduction

Today's automobiles are now sophisticated and connected software-controlled machines. However, attacks were originally made by accessing the CAN bus physically. Nevertheless,

subsequent studies found out that there are remote methods of exploiting the automobile system using Bluetooth, media, and telematics, which are vulnerable to cyberattacks (Valasek & Miller, 2015).

1.1 Introduction of the company

Formerly one of America's "Big Three" automobile companies, Fiat Chrysler Automobiles (FCA) extended its connected vehicle initiative with the Uconnect telematics system, which featured cellular capability, navigation, and emergency services for millions of cars. At the time, FCA had more than 200,000 employees in world and sold millions of automobiles each year across North America and Europe (FCA, 2014). [More Explanation](#)



Figure 2 : FCA

1.2 Introduction to Cyber Threats and Risks

[Click Here](#)

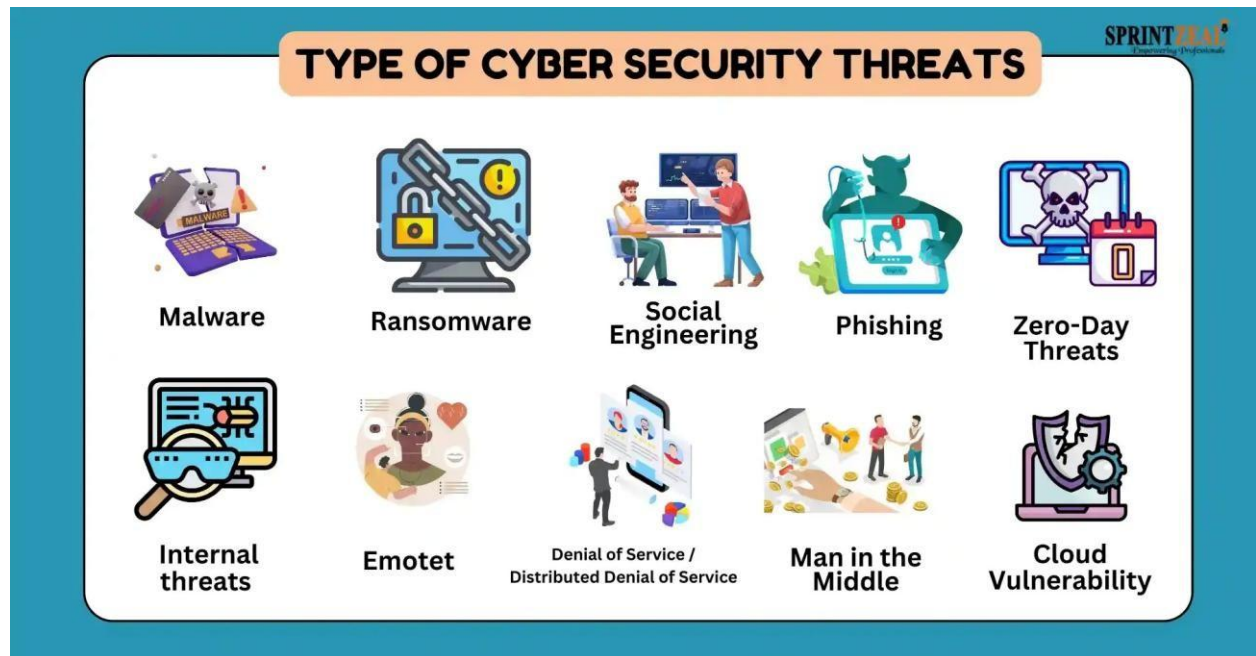


Figure 3 : Types of Cyber Security Threats

1.3 Aims and Objectives

[Aims and Objectives](#)

2. Background of the Incident

From 2010 to 2014, there were some major advancements made regarding the vulnerability of cars through tools that have been funded by DARPA. The researchers investigated whether CAN message injection on the steering wheel would impact car systems that use lane assistance and auto-parking. However, manufacturers claimed that such attacks needed physical access to vehicles (Valasek & Miller, 2015). [More](#)

2.1 Similar Case Studies

#	Incident	Year	Key Method	Impact
1	Nissan Leaf API Vulnerability	2016	Unauthenticated API access via VIN spoofing.	Remote control of climate systems and battery drainage; exposure of 2 years of driving history.
2	Mercedes-Benz E-Class Backend Hack	2019	eSIM analysis and unauthenticated backend requests via "Mercedes me" app.	Researchers gained remote access to unlock doors, flash lights, and start engines for ~2 million vehicles in China.
3	Toyota Supply Chain Ransomware	2022	LockBit ransomware targeting Kojima Industries (a Tier-1 supplier).	Forced shutdown of all 14 Japanese plants, resulting in a loss of 13,000 vehicles and ~\$375M in one day.
4	Honda "Rolling-Pwn" Attack	2022	Rolling code counter resynchronization (CVE-2021-46145).	Long-distance remote unlocking and engine start affecting most Honda models from 2012–2022.
5	Kia/Hyundai "TikTok Challenge"	2022	USB cable mechanical override on vehicles lacking engine immobilizers.	Massive increase in vehicle thefts (up to 9x in some cities), leading to a \$200M class-action settlement.
6	Mazda Infotainment USB Exploit	2017	Malicious script execution via the dashboard's USB port.	Allowed for unauthorized firmware modifications and the installation of custom, potentially malicious applications.

Figure 4: Similar Case Studies

2.1.1 Case Study 1: BMW ConnectedDrive Vulnerability (2015)

[Click Here](#)

2.1.2 Case Study 2: Tesla Model S Ethernet Breach (2015)

[Click Here](#)

2.2 The Wh Questions

Question	Answer
What happened?	Two security researchers remotely hijacked a Jeep Cherokee traveling at 70 mph on a public highway, gaining unauthorized control over non-safety functions (radio, AC, wipers) before graduating to safety-critical systems, such as the engine and brakes.
Who was involved?	Charlie Miller and Chris Valasek (security researchers), Andy Greenberg (Wired journalist/driver), Harman International (system developer), Fiat Chrysler Automobiles (vehicle manufacturer), and the NHTSA (federal regulator).
Where did it happen?	The live demonstration occurred on an elevated stretch of Interstate 64 in St. Louis, Missouri , while the researchers operated wirelessly from a basement ten miles away
When did it happen?	The technical exploit was privately shared with FCA in January 2015 , but the public demonstration and the subsequent recall of 1.4 million vehicles took place in July 2015
Why did it happen?	Structural architectural flaws allowed the internet-connected Uconnect system to communicate directly with the vehicle's internal CAN bus without network segmentation or "air-gapping" between the radio and safety controllers.
How did it happen?	The attackers exploited an unauthenticated open port (6667) via the vehicle's cellular connection, allowing them to rewrite firmware on the V850 chip and inject malicious commands onto the Controller Area Network.
What was the impact?	It led to the first-ever mass safety recall for cybersecurity defects , triggered a federal investigation into infotainment suppliers, and catalyzed the development of global standards like ISO/SAE 21434 .
What could have happened?	In the hands of a malicious actor, this vulnerability could have enabled mass-scale remote assassinations , coordinated traffic sabotage, or large-scale surveillance by tracking the location and audio of millions of drivers simultaneously.

Figure 5 : The WH Questions

2.3 System Failures That Enabled the Attack

Reasons

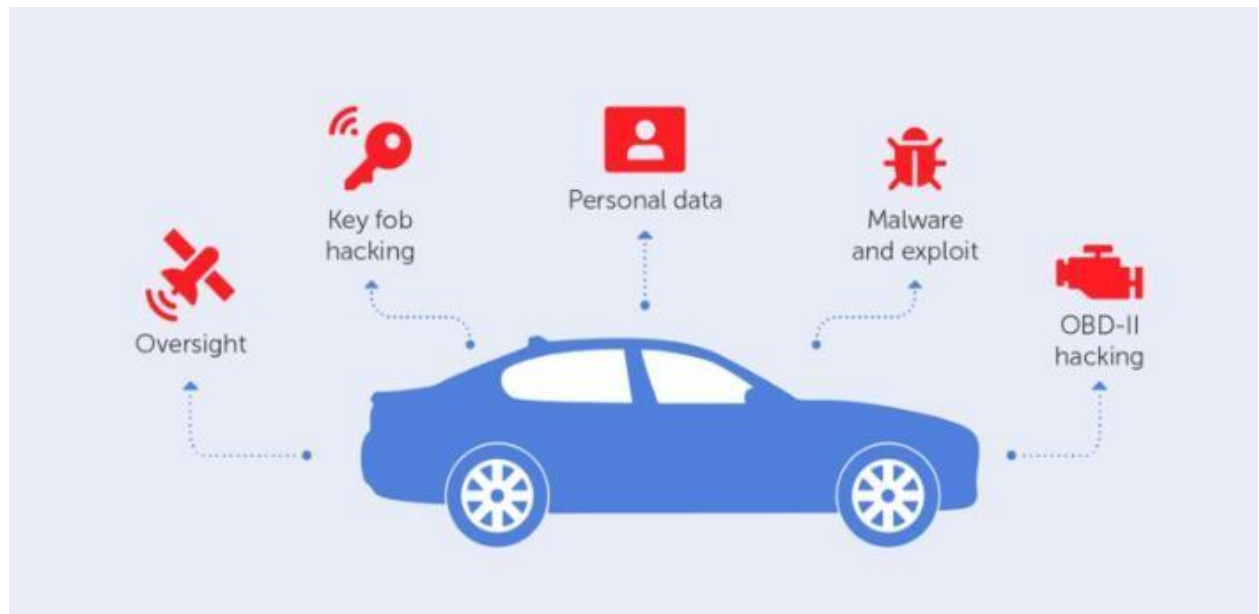


Figure 6 : Ways to hack Jeep Cherokee

2.4 Intention Behind the Unethical Act

[Explanation](#)

2.5 Impact if Done by Malicious Hackers

[Click Here](#)

3. Social Issues

Social issue refers to the impact of technology on society at large and individual behavior. It focuses on studying the impact of the event on the lives of individuals and their interactions.

3.1 Public Safety Concerns

Major social problem is threat to physical safety and human lives. In comparison to regular IT attacks where only the risk of damage to information/data/financial loss exists, the danger here is "cyber-physical systems," where digital commands lead to actions (Erwin, 2021).

- **The Weaponization of Transit:** The hacking showed that automobiles could be used as weapons through remote access. The scientists were able to disable safety features including steering, braking, and engine transmission while driving on highways speeds (Dro, 2015).

- **Sociotechnical Vulnerability:** In this case study, society was made aware of hidden dangers associated with technology. With greater vehicle connectivity, new threats are posed to road users and drivers that the system is ill-equipped to handle (Villasenor, 2015).

3.2 Trust in Connected Technologies

Jeep Hack Created "Crisis of Confidence" on the (IoT). For years, cars were considered "isolated castles" because of their mechanical nature and ability to protect passengers, but now they have become "connected endpoints" vulnerable to worldwide attacks.

- **Erosion of the Social Contract:** There exists an implied social contract whereby technology is supposed to improve safety. When a so-called smart feature undermines brake functionality, public confidence in such technology is seriously undermined (Fagan, et al., 2020).
- **Demographic Anxiety:** Research shows that public concern about (CAVs) is not uniform; demographics including age and location influence how much "safety anxiety" a person feels regarding remote hacking (Lee & Hess, 2022).

3.3 Impact on Society and Consumer Behaviour

The scandal caused a change in way customers look at the worth of a car, changing from viewing it based on "features" such as Wi-Fi and applications to "integrity" like security and updates.

- **Changes in Consumer Advocacy:** In the years following the hack (2015–2020), consumer advocacy groups began demanding slower deployment of connected features, calling for "Security-by-Design" as a mandatory standard (Lee & Hess, 2022).
- **Reputational Damage and Market Behavior:** However, high-profile cyber-attacks lower confidence in smart automobiles, prompting people to go for traditional technology, which is seemed more secure and less susceptible to such attacks.

3.4 Psychological Fear Among Vehicle Users

Psychological Impact	Detailed Explanation
Loss of Driver Agency	The drivers felt a sense of loss of control, as they came to understand that their commands through steering and braking could be interfered with by an unknown entity at a distance.
Connectivity Stigmatization	A high-profile breach results in the stigma associated with "smart" functions, which are considered a "Trojan horse" for the potential threats to safety. (White, 2023)
Autonomous Trust Deficit	The public fear of vulnerability in telematics prevents the acceptance and adoption of self-driving cars. (Villasenor, 2015)
Surveillance Anxiety	In addition to fears about physical safety, there is the fear that these same channels could also be used for continuous monitoring of their location and behaviour. (Moore, 2026)

Figure 7 : Psychological Fear Among Vehicle Users (White, 2023) (Villasenor, 2015) (Moore, 2026)

3.5 Public Awareness About Automotive Cybersecurity

Awareness Category	Detailed Explanation
Cyber-Safety Ratings	Cybersecurity is regarded as an important metric when purchasing a product; consumers research a brand's track record in implementing software patches with as much detail as their crash tests. (PWC, 2026)
Researcher Legitimacy	The perception of "white-hat" hackers has changed, from perceiving them as security risks to seeing them as necessary "digital whistleblowers." (PSU, 2018)
Transparency Demand	Consumer awareness and demand for "Software Bill of Materials" (SBOM) are increasing, calling for greater transparency about software and apps used in vehicles. (Fagan, et al., 2026)
Mass-Media Literacy	Coverage of exploits that can be described as "viral," like the one involving the Kia-Hyundai car theft incident, has taught ordinary consumers how dangerous software can be.

Figure 8 : Public Awareness about Automotive Cybersecurity (PWC, 2026) (PSU, 2018) (Fagan, et al., 2026)

4. Ethical Issues

Ethical concerns revolve around morality and the idea of right versus wrong. Ethics are different from law in that they tend to be subjective and pertain to the choice between two competing goods or evils within philosophical paradigms.

4.1 Ethical Responsibility of Hackers

Both Charlie Miller and Chris Valasek showed professional integrity by adhering to responsible disclosure, maintaining their white hat ethics to minimize any damage while also addressing vulnerabilities in manufacturers vehicles.

- **The Ethics of Risk-Taking:** In terms of utilitarianism, the live hacking session was justifiable for the protection of millions of automobiles and people. Nevertheless, deontological ethics does not support using uneducated people as a tool to gain wider benefits (Villasenor, 2015).
- **A Duty to Disclosure:** Virtue Ethics , as described by Miller and Valasek, was proved by their compliance with "Responsible Disclosure" by alerting Fiat Chrysler Automobiles months in advance to balance between honesty & safety for the people (DUPLCHIN, 2017).

4.2 Corporate Responsibility and Negligence

The underlying principle that makes the scandal unethical centres around negligent design. The duty of care of both Fiat Chrysler, and their vendors was to produce something that is safely designed.

- **Failure of Due Diligence:** According to **Kantian Deontology**, organizations have a “categorical imperative” to make decisions based on rules that safeguard human lives. In this case, by not ensuring the security of Port 6667, FCA violated their obligation to ensure the safety of the consumers who were their primary customers (Higgins, 2015).
- **The "Safety vs. Profit" Dilemma:** The case was a breach of the social contract as consumers anticipated safety in the vehicles. The prioritization of connective features over building a secure product was a failure on the side of manufacturers (King & Dan, 2016).

4.3 Privacy and Data Protection Concerns

The hack proved that a breach of security is also a breach of **Privacy**. Vehicles are no longer just transportation; they are "data centers on wheels."

- **Non-Consensual Surveillance:** The act of tracking a car's GPS and Velocity secretly is unethical because it breaches the Right to Privacy. It is unethical that a "connected" car should be used as a spy device without one's consent (Dro, 2015).
- **Data Integrity as a Human Right:** With the storage of increasing amounts of personal data on board, vehicle manufacturers become increasingly ethically responsible for protecting their consumers from any form of breach of their privacy (King & Dan, 2016).

4.4 Responsible Disclosure of Vulnerabilities

Automobile companies have developed an ethical understanding with their researchers known as responsible disclosure to address any vulnerabilities before making any public announcements. This practice came into being due to the hack of a Jeep in 2015 (Bolz & Kriesten, 2021).

Feature	Ethical Theory Connection	Detailed Ethical and Professional Explanation
Standardized Reporting	Rule Utilitarianism	ISO/SAE 21434 offers an approach where automakers have a process-based approach to handling vulnerability reports in their cars through their entire life cycle.
Bug Bounty Adoption	Incentive-Based Ethics	In response to what happened in 2015, several automobile manufacturers came up with programs called "bug bounty," which rewarded people for disclosing problems instead of taking advantage of them (Bolz & Kriesten, 2021).
Coordinated Patching	Duty of Care	With responsible disclosure, manufacturers are required to have enough time to address the problem with fixes by using OTA before any technical information is revealed (Bolz & Kriesten, 2021).

Figure 9 : Responsible Disclosure of Vulnerabilities (Bolz & Kriesten, 2021)

4.5 Ethical Conflict Between Innovation and Safety

Ethical Conflict	Ethical Theory Connection	Detailed Explanation of the Dilemma
Connectivity vs. Security	Teleological Conflict	Adding smart functionality to the car through cellular connectivity rapidly expands its vulnerability profile by embedding backdoors, which designers may be oblivious to.
Algorithmic Opacity	Right to Explainability	AI-assisted decisions in cars are typically opaque and lack a clear "right to explainability," thus failing to ensure users' confidence in their reaction in critical situations (Gherardini & Cabri, 2024)
Liability Shifts	Accountability Ethics	The ethical responsibility for the safety of the car moves from the driver to the car manufacturer, and finally to software vendors when cars become more autonomous. (Bozdal , et al., 2018)

Figure 10 : Ethical Conflict Between Innovation and Safety (Gherardini & Cabri, 2024) (Bozdal , et al., 2018)

#	Ethical Issue	Utilitarianism	Deontology	Virtue Ethics	Rights-Based Ethics
4.1	Ethical Responsibility of Hackers	The live hacking session was justifiable for the protection of millions of automobiles and people (Villasenor, 2015).	Does not support using uneducated people (the public) as a tool to gain wider benefits (Villasenor, 2015).	Balancing honesty and safety through "Responsible Disclosure" alerts (DUPLECHIN, 2017).	The public's right to safety was prioritized through the exposure of fatal flaws.
4.2	Corporate Responsibility & Negligence	Prioritizing connectivity features over security-by-design contradicts the safety of the majority.	FCA violated the categorical imperative to safeguard human lives at Port 6667 (Higgins, 2015).	An ethical organization fulfills its "Duty of Care" by producing safely designed products.	Consumers bought cars on the understanding that the producer would keep them safe.
4.3	Privacy & Data Protection Concerns	Breach of security is a breach of privacy; protecting the "data center on wheels" is a utility.	Non-consensual tracking of GPS and velocity is an unethical breach of the Right to Privacy (Dro, 2015).	Manufacturers are ethically responsible for protecting consumers from any form of data breach.	Data integrity and privacy are treated as fundamental human rights for vehicle users.
4.4	Responsible Disclosure of Vulnerabilities	An ethical agreement to address gaps before public announcement maximizes global safety.	Manufacturers have a duty to allow researchers to disclose bugs through formal programs.	Compliance with disclosure timelines shows integrity and balances honesty with public safety.	The public has an inherent right to be informed of vulnerabilities before they are exploited.
4.5	Innovation vs. Safety Conflict	Expanding functionality via cellular connectivity embeds backdoors that oblivious designers may miss.	International standards (ISO) now mandate a legal duty to patch gaps before revealing them.	Rigorous standards (white-hat outreach) demonstrate a professional commitment to safety.	The user's "Right to Explainability" in AI-assisted decisions is a core ethical right.

Figure 11 : comparing ethical issue with 4 theories.

5. Legal Issues

Legal problems refer to the use of statutes, policies, and judicial decisions. It is the strictest field since it entails rules and penalties such as monetary fines or incarceration for offenses committed by individuals.

5.1 Computer Fraud and Abuse Act (CFAA) - 18 U.S.C. § 1030

CFAA is the central legislation regarding unlawful access to computer networks in the U.S. The offense of willful impairment of computer operations **under 18 U.S.C. § 1030(a)(5)** is a criminal act (Congress, 2020). It must be noted that despite Miller and Valasek being white-hat researchers,

the CFAA does not make any distinction regarding malicious intentions or security studies. This event prompted the U.S. Copyright Office to provide the DMCA Exemptions for Section 1201 (Offcie & Congress, 2015).

5.2 National Traffic and Motor Vehicle Safety Act - 49 U.S.C. § 30101

This law confers on the NHTSA the mandate of regulating vehicle safety. **49 U.S.C. § 30118** states that manufacturers are obligated by law to report owners about "safety related defects" (Congress, 2021). In an unprecedented decision following the Jeep hack incident, NHTSA came up with its Guidance **Enforcement Bulletin 2016-02**, which recognizes cybersecurity flaws as safety defects (NHTSA, 2016). For this reason, FCA was fined \$105 million under **49 U.S.C. § 30165** for compromising consumer safety (Congress, 2015).

5.3 Wire Fraud Statute - 18 U.S.C. § 1343

Sending the fraudulent commands via the Sprint cellular network with an intent to alter the operation of the vehicle's computer system is also wire fraud as per Section 18 U.S.C. § 1343. that states that "any fraudulent scheme" is forbidden to be carried out with wire communication across state lines. This law provides up to twenty years of imprisonment per charge (congress, 2023).

5.4 Privacy and the Electronic Communications Privacy Act (ECPA)

GPS tracking and speed tracking without authorization constitute a breach of the Right to Privacy. The **Electronic Communications Privacy Act under 18 U.S.C. § 2510** makes it illegal for anyone to intercept communications that are electronically transmitted (Congress, 1968). As "connected" cars operate like mobile data centers, monitoring of them in any way without consent amounts to breach of **Stored Communications Act under 18 U.S.C. § 2701** (Congress, 2018).

5.5 Global Standards: UN Regulation No. 155 (UN R155)

In the international arena, this incident resulted in the formation of **UN Regulation 155**, which requires vehicle makers to adopt a CSMS that has been certified. Vehicles must pass "Type Approval" tests to prove they have security measures that protect them against any remote attacks during their entire life cycle. The Uconnect platform of 2015 would not be allowed on the market under this regulation (UNECE, 2021).

6. Professional Issues

Professional concerns pertain exclusively to those relating to the principles, code of ethics, and obligations of specialists in a certain field.

6.1 Software Engineering Responsibility

Automotive software engineers must ensure connectivity does not threaten safety. The Jeep hack revealed poor system segregation, allowing critical vehicle systems to remain vulnerable to cyberattacks

- **System Integrity and Isolation:** Professional ethical codes like those of ACM and BCS mandate that engineers do not cause any harm. The lack of segregation of the infotainment system from the CAN bus is an indication of negligence on the part of engineers (BCS, 2022).
- **The Shift to "Secure Software Development Lifecycle" (SSDLC):** After the event, car software engineering moved towards Security by Design, which means that security had to be incorporated in cars right from the start and could not be applied retrospectively (Sherman & Palluch, 2015).

6.2 Industry Standards and Best Practices

The cyberattack on Tesla in 2015 highlighted the absence of uniform standards for cybersecurity in the automotive sector. To address this issue, certain prominent professional codes have been introduced to regulate engineers' work (Erwin, 2021).

- **ISO/SAE 21434:** This is presently the most important global standard in relation to “Road Vehicles Cybersecurity Engineering.” It provides a professional basis for ensuring the management of cybersecurity threats from the initial design phase to decommissioning of a vehicle.
- **SAE J3061 :** Formulated by SAE’s Vehicle Cybersecurity Committee, the framework served as the starting point for ISO 21434, where cybersecurity is viewed as a component of vehicle functional safety and not merely an information technology issue (Team, 2020).

6.3 Risk Management and Accountability

The Jeep hack revealed that engineers working in the automotive industry should adopt systematic approaches to risk management because it is no longer acceptable for professionals to overlook unintended system interactions. (Acsia, 2024).

- **Threat Analysis and Risk Assessment:** Use of TARA to systematically assess what an asset is (such as an ECU) along with assessing the feasibility and implications of cyberattack on the asset has become one of the core practices of professionalism today (Camacho, 2021).
- **Accountability in the Supply Chain:** In the Jeep attack, third-party software modules were used, indicating that there is a need for shared responsibility when it comes to cybersecurity (Erwin, 2021).

6.4 Importance of Security Testing and Maintenance

Key Area	Technical and Professional Explanation
Continuous Lifecycle Monitoring	According to the ISO/SAE 21434 standard, cybersecurity management must be done from design all the way through decommissioning. Vehicle testing should not just happen in the factory, but also during its 10 to 15-year life span to counteract any new “zero-day” threats. (Embitel, 2026)
Dynamic Penetration Testing	Whereas static analysis checks the design of the code, dynamic analysis entails simulated attacks (such as fuzzing and signal spoofing) on the code during actual vehicle use.
Vulnerability Management	Cyber Security regulations, such as those set by UNECE R155, require an OEM to have a process for identifying, tracking, and mitigating vulnerabilities in vehicles through a CSMS. (Bildirici, 2024)
Secure OTA Updates	Modern maintenance depends on OTA. Authenticity and integrity of these software OTA updates are very important to avoid 'mass-malicious updates' affecting a whole fleet of vehicles at once.

Figure 12 : Importance of Security Testing and Maintenance (Embitel, 2026) (Bildirici, 2024)

6.5 Professional Ethics in Cybersecurity Practices

Professional Issue	Code of Conduct Reference	Detailed Professional Explanation
Violation of Public Welfare	ACM 1.1 / IEEE 1	Professionals have a fundamental duty to advance the welfare of humanity. Applying technical expertise to sabotage the brakes on an operating automobile is an act that clearly contravenes the dictum to "hold paramount the safety of the public."
Abuse of Specialized Knowledge	IEEE 2 / BCS 1.1	Honesty regarding potential hazards is expected from all professionals. The identification of an unsecured port (6667) and its exploitation through sabotage, rather than fixing, would amount to a serious misuse of "specialized power" vested in engineers.
Failure of System Integrity	ACM 2.1 / SE Code 3.01	Software developers must make sure that the software they develop adheres to the highest professional standards. It is unprofessional to ignore the absence of airgaps between the entertainment system and the safety system.
Breach of Confidentiality / Privacy	ACM 1.7 / BCS 2.1	The secret access of the driver's GPS location and speed constitutes an infringement on the professional obligation of "honoring confidentiality"
Duty to Report (Whistleblowing)	ACM 1.2 / BCS 1.3	If a professional finds a potentially fatal problem but finds that he is not being heeded by management, then he has an obligation to raise the red flag, and neglecting to do so would make him culpable for any damage that results.

Figure 13 : Professional Ethics in Cybersecurity Practices (ACM, 2018) (BCS, 2022)

7. Conclusion and Personal Reflection

The Jeep Cherokee attack is undoubtedly "watershed moment" in auto-cybersecurity, representing the shift from potential digital attacks to real physical risks.

7.1 Summary of Key Findings

[Click Here](#)

7.2 Ethical Decision-Making Methodology (Brainstorming & Analysis)

To make sense of the 2015 Jeep case study, I have used the Methodology for Ethical DecisionMaking, emphasizing the brainstorming and analysis stages.

A. Brainstorming Phase (Identifying Alternatives)

At this stage, I outlined three possible routes that the stakeholders (the engineers/researchers) may have chosen to take:

1. **Path Alpha (Complete Secrecy):** The researchers discover the bug but sell it in the black market or keep it secret, putting the 1.4 million drivers in jeopardy.
2. **Beta Path (Subtle Patching):** The researchers privately inform FCA, and FCA subtly patches the new vehicles, leaving the old vehicles in their unpatched state to prevent public outrage.
3. **Gamma Path (Responsible Disclosure):** The approach taken where the vendor is notified, given a time limit, and the public is informed (ACM, 2018).

B. Analysis Phase (Evaluating the Ethical Impact)

- **Impact on Stakeholders:** Path Alpha creates a higher risk of “unjustified physical injury” (ACM 1.2). Path Beta gives priority to corporate gains over the “Public Good.”
- **Rights and Responsibilities:** In accordance with the Social Contract, the public has a right to be made aware of fatal flaws. Only Path Gamma satisfies the “Duty of Care (BCS, 2022).
- **Solution:** Path Gamma has been chosen as the most ethical choice because it achieves maximum transparency and compels safety-related policy changes at a higher cost for the manufacturer.

7.3 Personal Reflections and Recommendations

From my experience as an upcoming IT specialist, personal reflections in this case should focus on Ethical Dissent. The hack attack on the Jeep highlights the importance of preventing “Connectivity” from becoming the enemy of “Safety.”

- **Recommendation 1 (Technical):** It is highly recommended that “air-gapping” of infotainment from the CAN bus system be enforced legally for all smart vehicles to protect against the lateral transfer of malicious attacks.
- **Recommendation 2 (Professional):** Engineers must be able to “blow the whistle” when they find a vulnerability in life-critical systems. Security is not an IT problem it’s a human rights problem.
- **Recommendation 3 (Governance):** In my opinion, the automotive manufacturer should be required to keep a SBOM (Software Bill of Materials) for all vehicles, which will help to detect any vulnerabilities once any cybersecurity threat or CVE comes out.
- **Recommendation 4 (Consumer Empowerment):** I would recommend an independent cyber safety rating system for automobiles to evaluate the level of cybersecurity standards, increase consumer consciousness, and prompt manufacturers to focus on cybersecurity along with automobile efficiency.

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9. Appendices

Appendix A: Introduction to FCA

The Jeep Cherokee was among its many popular vehicles, which came outfitted with a connected infotainment system referred to as Uconnect, a joint venture between Fiat Chrysler and Harman International Industries that featured GPS navigation, phone connectivity, and remote vehicle monitoring through Sprint cellular technology. By the time of the security breach incident, Uconnect had been implemented in about 1.4 million FCA automobiles across America (BBC, 2015).

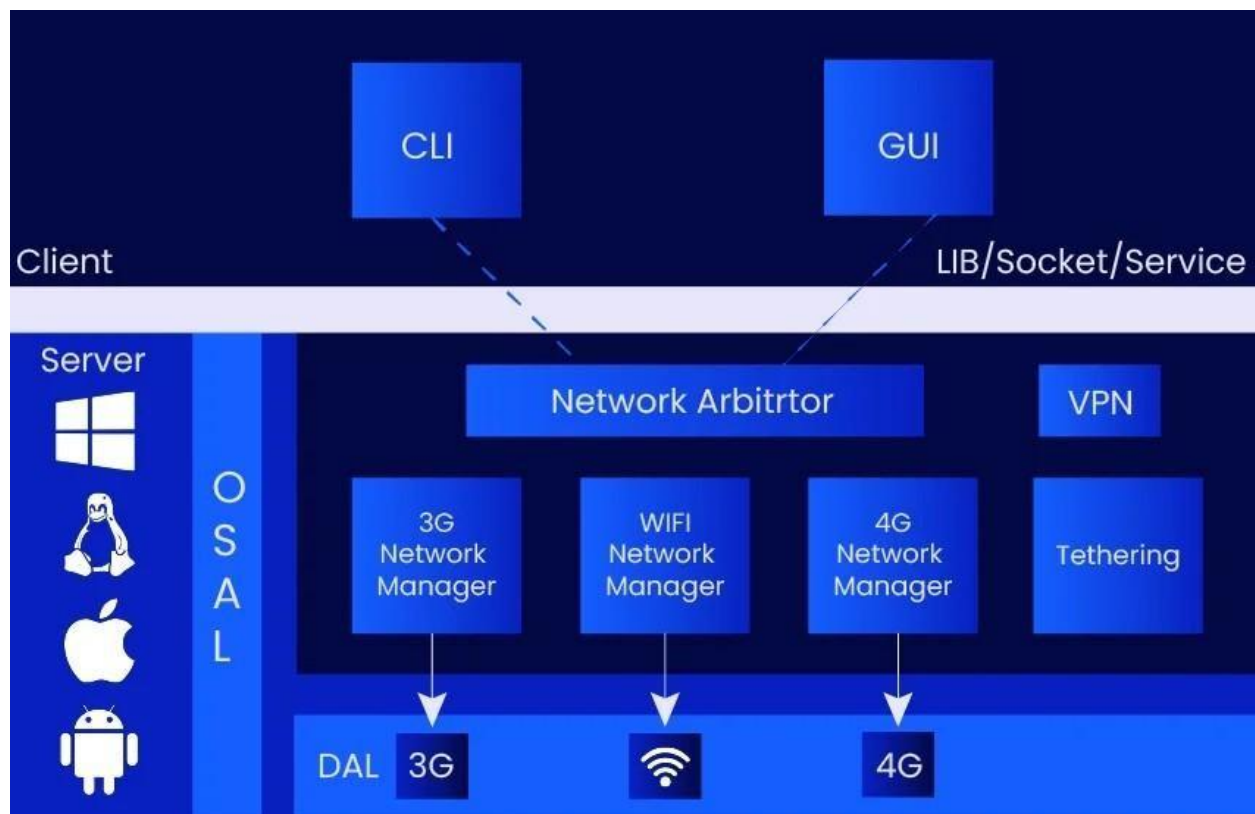


Figure 14 : Illustration of Uconnect System

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Appendix B: Introduction to Cyber Threats and Risks

There has been a paradigm shift in automobile industry such that cars are no longer considered machines, but rather as data centers on wheels. The consequence of this shift is that there have

emerged new kinds of cyber-physical threats whereby the cyber threat could translate into something tangible.

For the present era of 2026, there are three major risks associated with connected vehicles:

- **Risk of Remote-Control Compromise:** In the Jeep Cherokee example, the greatest risk is hacking through firewalls to transmit malware to the Controller Area Network (CAN bus) that can disable brakes or kill the engine on highways
- **Risk to Privacy & Data Collection:** Today, a car collects immense telemetry, GPS, and even biometric data. A telematics system that does not follow security protocols results in a risk called "Unauthorized Surveillance" where a driver can be located and monitored without his knowledge
- **Vehicle Supply Chain and API Security Threats:** Vehicle functionality has been outsourced to third-party applications and cloud-based APIs, leading to a "Broad Attack Surface." If any telematics service provider has a security flaw, it will affect the entire fleet of vehicles simultaneously.

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Appendix C: Aims and Objectives

The main purpose of this report is to provide an in-depth analysis of the 2015 Jeep Cherokee hack to determine the multi-dimensional consequences of the security threats posed by the cyber

vulnerabilities that exist in today's connected automobiles. By studying this milestone event, this report attempts to promote the "Security-by-Design" concept in the automobile industry.

The specific **objectives** required to achieve this aim are:

- For studying the technical vulnerabilities of the car in terms of unauthenticated remote access through Port 6667 and lack of network segmentation (air-gapping) that made remote hijacking of critical vehicular functions possible.
- For identifying the ethical issues raised by the case in terms of comparing "White Hat" responsible disclosure by the hackers against corporate irresponsibility shown by Fiat Chrysler Automobiles (FCA).
- For determining the social repercussions of connected car breaches, particularly considering the damage done to consumer trust in IoT technologies and fear related to "cyber-physical" attacks on individuals.
- For identifying the legislative changes that followed because of the hack, especially with the enactment of the SPY Car Act, UN Regulation 155, and DMCA changes.
- To set professional obligations for software engineers and producers, focusing on the importance of compliance with international guidelines such as ISO/SAE 21434 and having adequate risk management systems.
- To generate recommendations for addressing cybersecurity issues in the automotive industry through Zero Trust Architecture and bug bounty initiatives. [Back to 1.3](#)

Appendix D: Extended Background of Incident

All these assumptions were challenged when researchers chose to study the Jeep Cherokee because of its high level of connectivity along with relatively easy-to-exploit architecture. These specialists sought to prove that not only was the possibility of remotely attacking an unchanged car plausible, but it also entailed certain danger. In doing so, they managed to successfully exploit vulnerability

to gain remote access and control key vehicle functionalities. It opened a new page in terms of research as well as raised a whole lot of concerns regarding attention (Valasek & Miller, 2015).

The specific target for the attack was the Harman Uconnect infotainment system, which they found acted as an unintentional entry point into the computer system inside the vehicle. By taking advantage of an unauthenticated D-Bus service running at Port 6667, the researchers showed that they were able to remotely run malicious code using the Sprint cellular network. Through this entry point, they were able to move from the “non-safety” system to the V850 controller (Valasek & Miller, 2015).

The implications of this exploit are that it completely invalidates the concept of Security through Obscurity within the automotive industry. FCA and other manufacturers had previously felt that the use of CAN protocol was enough protection against remote attacks because of the level of complexity involved. Once lateral movements were made in the network, however, the researchers were able to gain access to mission-critical systems, such as transmission, steering, and braking (Valasek & Miller, 2015).

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Appendix E : Case Study 1: BMW ConnectedDrive Vulnerability (2015)

Early in 2015, the Allgemeiner Deutscher Automobile-Club (ADAC) of Germany revealed that there was a vulnerability in the BMW's ConnectedDrive system, which is a telematics service provided by BMW. It was found that the system communicated using unsecured transmission through the mobile network SIM card in the vehicle; an attacker could pretend to be a mobile network tower and impersonate a legitimate BMW server and therefore take control of the wireless

connection. Using vulnerability, an attacker could unlock car doors, retrieve the location of the vehicle, and use other comfort services offered by the system (Region-I, 2015).

In response to this issue, BMW rolled out an OTA software update that would encrypt communications between the cars and the backend server, thus closing the loophole by January 2015. While the bug didn't provide access to critical vehicular systems like steering or brakes, it showed that cellular-based telematics gateways were susceptible to being bypassed to carry out attacks against other vehicular functionalities (Williams, 2015).

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Appendix F : Case Study 2: Tesla Model S Breach (2015)

During the Def Con 23 event held in August 2015, cybersecurity experts such as Marc Rogers (Cloudflare) and Kevin Mahaffey (Lookout) conducted a complex multistep hacking demonstration on the Tesla Model S, revealing that even highly secured electric cars may be hacked if the hacker is successful at obtaining physical access first. Cybersecurity researchers took apart different components of the car interior until they identified an Ethernet cable that was connected to the internal network of the automobile. They plugged in a computer, and using the software flaws of the vehicle's infotainment and CAN-bus-based systems, they were able to inject backdoor Trojan, giving them remote access to many of its non-critical systems (Goodwin, 2015).

After implantation, the hackers were able to remotely unlock doors, switch on/off the vehicle, alter the reading of the speedometer, and control various amenities, such as the windows and suspension, allowing them to park the vehicle by taking it to a stop at slow speeds while undergoing a test drive. The company quickly resolved the problem through over-the-air software updates. This instance is important in history because it involved intrusion through physical access through the Ethernet port followed by remote control. It differs from the Jeep Cherokee incident, where hackers achieved remote control without any physical access in a publicly available car over cellular networks for the first time (Goodwin, 2015). [Back to 2.1.2](#)

Appendix G: System Failures That Enabled the Attack

The below-mentioned five systems failures played an important role in helping the scientists get remote control over the vehicle:

- **Unauthenticated Remote Access through Internet Port 6667:** The Harman Uconnect infotainment system exposed an unauthenticated, public internet port (Port 6667) that operated by D-Bus system inter-process communication, meaning there was no requirement for any login credentials or encryption. Any device connected on the same cellular network could discover and gain remote access to the vehicle (CISA, 2018).
- **Failure of Network Segmentation (Air-Gapping):** A critical failure was the lack of segmentation between the entertainment systems and other safety modules within the car. Since the radio was linked to the CAN bus, there was no "air gap" between the entertainment system and other modules in the vehicle, meaning that once the entertainment system was compromised, full access could be gained to other critical modules controlling the brakes, wheels, and transmissions (Bozdal , et al., 2018).
- **Firmware Integrity Assurance Failure:** There was also a failure of firmware integrity in the sense that unauthenticated firmware updates were supported by the system. Malicious hackers could install new firmware into the entertainment system without any need for verification. A "code signature" was not required to prove the authenticity of the installed update (Garlati, 2015).
- **Insecure CAN Bus Protocol Design:** As for the CAN bus protocol, it was implemented by default in the vehicle, and the protocol does not include any type of encryption and sender identification. This protocol is based on "broadcast," meaning that any node is allowed to send a request to another node, thus allowing the hackers to forge commands once they got inside the "gateway" (Bozdal , et al., 2018).
- **The Lack of External Port Screening:** From the viewpoint of the network provider, Sprint failed to screen the traffic directed to the vulnerable port number 6667. It means that no external protection measure was taken to safeguard the vehicle against the attack coming from the entire internet domain, and not even a VPN could protect the vehicle (CISA, 2018).

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Appendix H: Intention Behind the Unethical Act

Although the experiment entailed tampering with an automobile that was in use on a real-life highway an activity that entails certain risks the fundamental motivations were driven by a welldefined ethical code:

- **Responsible Disclosure and Preventive Security:** In accordance with established industry protocols, the authors conducted a private briefing with the manufacturer, Fiat Chrysler Automobiles (FCA), nine months prior to their presentation at the conference. Through what is known as “responsible disclosure,” they allowed the company to take proactive measures to create a security patch well in advance of the publication of these results (Erwin, 2021).
- **Forcing Industry Accountability:** Another goal of the researchers' unethical breach was to force the automotive industry to take action to address its “air gap” mentality and acknowledge the necessity of investing in cybersecurity. They were successful in convincing upper management to address their vulnerabilities and assign a team solely responsible for conducting tests for potential security flaws in products (Villasenor, 2015).
- **Public Awareness and Policy Reform:** Another objective of theirs was to trigger regulation. The subsequent massive recall of one and a half million automobiles along with the establishment of the SPY Car Act is evidence enough that their “ethical breach” was required to trigger legal reform that enables organizations such as NHTSA to ensure safe driving through data security of cyber-physical systems (Erwin, 2021).
- **Unreasonable Risk Validation:** It was intended that this study would establish through empirical data that unauthenticated open ports on automobiles today represent an "unreasonable risk" to human safety. This involved making sure the automobile industry takes steps to ensure that all commands issued in the automobile network are constantly validated under a Zero Trust Architecture framework (Erwin, 2021).

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Appendix I: Impact if Done by Malicious Hackers

- **Public Safety Hazard Causing Large Numbers of Casualties:** A malicious actor could leverage the cellular “find” function to discover thousands of susceptible vehicles by targeting their IP addresses at once. The "broadcast" function could then be used to take

remote control of vehicle functions like engine stop and brake activation at highway speeds causing multiple crashes on busy roads during rush hour traffic and fatalities in the process.

- **Installation of Ransomware onto Vehicle Control Units:** If the hack could rewrite firmware, malware and even ransomware could be installed onto the vehicle's onboard computer system. This would enable the attacker to render useless or "brick" the ignition and transmission units in exchange for cryptocurrency payments before the attack can be neutralized.
- **Political or Corporate Assassination using Hack:** Due to the precision of GPS targeting and ability to control critical vehicle functions like steering and breaking, a hacker could use this exploit for politically or professionally motivated murders of individuals. The targeted person could be killed during a car crash orchestrated on a tight corner or suspension bridge.
- **National Security and Disruption:** State-backed players could conduct their own cyberphysical attacks through disrupting thousands of delivery trucks and emergency vehicles at once. By doing so, they could cripple the economy of a country, its emergency response systems, and cause civil unrest.
- **Crime and Avoidance of Prosecution:** Since cyber manipulation is not a tangible attack method, there will be little trace left for the police to follow up on the crime committed. Furthermore, criminals might take advantage of the possibility to use such an anonymous attack to commit any crime and get away from it because investigators will find it hard to differentiate between a software bug and a driver error.

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Appendix J: Summary of Key Findings

The investigation into this scenario yields several critical conclusions:

- **Architectural Failure:** The main technical reason for the vulnerability was the absence of the separation method known as "air-gapping," between the mobile networked

infotainment system and the vehicle's critical CAN (Controller Area Network) bus. (Villasenor, 2015)

- **Regulatory Evolution:** This occurrence necessitated a change in the legal framework, which led to the enactment of the SPY Car Act and the categorization of cyber security weaknesses under the "Safety Defects" of NHTSA. (NHTSA, 2016)
- **Professional Standards:** This pointed to the need for a competency shortfall in conventional automotive engineering, making it necessary to introduce new global standards like ISO/SAE 21434 for cybersecurity engineering.

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Appendix

K: Timeline of the Jeep Cherokee Hack

The timeline below describes the process from the beginning of an internal security research study to the issuance of a worldwide product safety recall.

❓ **Late 2013:** Charlie Miller and Chris Valasek begin researching automotive security, specifically targeting the Harman "Uconnect" system due to its widespread use in Fiat Chrysler Automobiles (FCA).

❓ **Early 2014 – Mid 2015:** The researchers identify that the Uconnect system's cellular connection is not properly isolated from the vehicle's internal Controller Area Network (CAN bus).

❓ **January 2015:** Acting as "White Hat" hackers, the researchers privately share a detailed technical briefing of the vulnerability with FCA. This allows the manufacturer a six-month head start to develop a remedy before the flaw is publicized .

❓ **July 16, 2015:** FCA releases a voluntary security patch; however, it requires customers to download the software to a USB drive or visit a dealership, leading to slow adoption.

❓ **July 21, 2015:** *Wired* magazine publishes a feature demonstrating the hack. The researchers remotely disable the transmission and brakes of a Jeep on a live highway with a journalist inside .

❓ **July 24, 2015:** Following intense pressure from the National Highway Traffic Safety Administration (NHTSA), FCA officially recalls 1.4 million vehicles to perform the software update .

Figure 15 : Detailed Timeline of the Jeep Cherokee Hack

Appendix

L: Technical Overview of the Uconnect Vulnerability

Vulnerability was due to a poor system design that had no proper network segmentation.

- **The Entry Point (Cellular Gateway):** The infotainment system had a Sierra Wireless cellular chipset installed in it. The researchers found out that Port 6667 (D-Bus) was exposed and unauthenticated. It meant that any device on the Sprint cellular network could "locate" and communicate with the car's radio system. (Valasek & Miller, 2015)
- **Privilege Escalation:** After gaining access to the radio system running the Linux OS, the researchers managed to initiate a firmware update in the radio's operating system. They succeeded due to the absence of "code signing" (procedure that guarantees the authenticity of the software installed), and as a result, they managed to inject malware into the system. (Dro, 2015)
- **The Bridge to the CAN Bus:** Radio Processor of Jeep Cherokee was connected to the V850 Controller responsible for the management of the CAN bus. Lack of "air-gapping" made it possible for researchers to send spoofing commands to the steering, braking, and engine modules thus bypassing user interaction with the car (Valasek & Miller, 2015)

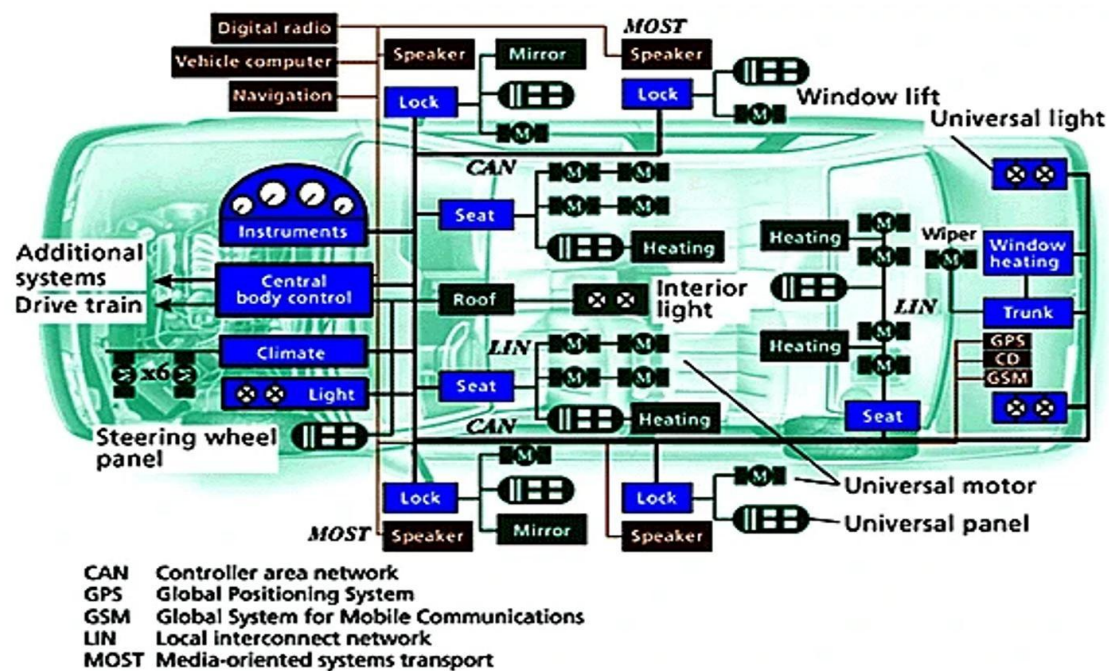


Figure 16 : Diagram of Vehicle CAN BUS Architecture

Appendix

M: Summary of Fiat Chrysler Recall

NHTSA Recall Number: 15V461000 was the first recall in which an automotive manufacturer recalled a large batch of automobiles due to a cyber issue rather than a mechanical one.

- **Affected Vehicles:** The recall extended to many types of vehicles, such as the Ram 1500/3500 (2013–2015), the Dodge Viper, the Jeep Cherokee, and the Chrysler 200/300 with an 8.4 touchscreen system
- **The Multi-Layered Remedy:**
 - Network Level:** With help from Sprint, FCA prevented remote access by blocking port 6667 at the cellular network level, prior to fixing cars with software updates.
 - Software Level:** An update for firmware was delivered that could enhance the “isolation” of the infotainment system from the car’s network systems.
- **Regulatory Impact:** The recall demonstrated that the NHTSA possessed the legal right to enforce recalls in the case of software defects that could endanger human life, establishing a precedence for all future "Connected Vehicles."

Appendix

Appendix N: Comparing Similar Automotive Cyberattacks

COMPARATIVE ANALYSIS: LANDMARK AUTOMOTIVE CYBERATTACKS				
FEATURE	2015 JEEP CHEROKEE	2015 TESLA MODEL S	2013 TOYOTA PRIUS	2013 FORD ESCAPE
 Primary Vulnerability	Unauthenticated public Port 6667 in Uconnect	Ethernet port and browser vulnerabilities	Lack of message authentication on the CAN bus	Lack of message authentication on the CAN bus
 Attack Vector	 Remote (Cellular Network/Sprint)	 Local/Physical (Ethernet connection)	 Local/Physical (Direct OBD-II port access)	 Local/Physical (Direct OBD-II port access)
 Critical System Impact	Steering, braking, engine transmission, and GPS	Infotainment control and ability to shut down engine	Steering and braking manipulation	Steering and braking manipulation
 Network Isolation	None (Radio linked directly to CAN bus)	Partial (Gateway used, but bypassed via Ethernet)	None (Standard broadcast CAN protocol)	None (Standard broadcast CAN protocol)
 Manufacturer Response	 Mass recall of 1.4 million vehicles; software patch	 Over-the-Air (OTA) software update	 Initially dismissed as "physically dependent" risk	 Initially dismissed as "physically dependent" risk
 Regulatory Impact	Sparked the SPY Car Act and UN R155	Advanced the use of Bug Bounty programs	Increased DARPA funding for automotive security	Increased DARPA funding for automotive security

Figure 17 : Comparing Similar Automotive Cyberattacks

Appendix O: Key Cybersecurity Terms and Definitions



Figure 18 : Key Cybersecurity Terms and Definitions